as vrea te rog sa imi faci un fisier python in care sa iei tot ce vezi aici pas cu pas si sa incerci sa imi faci un sablon pe care pot lucra si sa fac teste etc sa verific anumite lucruri dar fa l sa contina tot ce am avea nevoie

In a quantum ml scenario, i have raw data about credit card fraud and i want to reduce the complexity of the quantum realm as much as possible and efficientasize it. i am currently using a 4 qubit light cone method with a reduction to about 24 expressions from 32. Do a deep search about any scientific research on the domain, make it detailed and dont stop researching until u find a better variant to the one we are using. i want it to be thoughtfull and effective.

vreau sa faci un research amanuntit ia tot timpul de care ai nevoie pentru a gasi cele mai valoroase si nisate surse care ne pot ajuta sa implementam o idee diferita de turma care sa ne scoata in evidenta. cantareste fiecare optiune si plusurile si minusurile in

implementarea acestora si ofera-mi idei extra pentru a implementa si a realiza un algoritm eficient si ieftin care sa dovedeasca ca avem idei originale dar care aduc ceva proiectului. vreau sa fii sigur pe raspunsuri si sa nu te opresti pana nu gasesti minim 2 idei posibile implementarii si imbunatatirii in vreun fel a proiectului.

would another strategy for the ga in terms of generation count or elite count make for a better output by any means?

I have found an interesting idea regarding genetic algorithms, what can you tell me about them, what are the advantages and disadvantages and what would you recommend for an easier implementation

fa un recap la toate tpoate toate testele si configuratiile prin care am trecut ca sa pot documenta si pentru a asigura trasabilitae

I have a really really really really small attention span. can you tell me if the results are promising or not vs baseline?

analyze all three notebooks (lightcone, task2 rank safe and task2 scion) and come up with a good implementation plan if it is possible and you think it can improve the results

make the task1 etc usable for cuda devices, especially the 11th task

ok for the task 11 of the ligthcone, I can implement as input layer preprocessing any type of layers. Do you think a lipschitz constraint denoising method would help? If so try to do a normal ml layering preprocessing

but what if we add a cnn and a constraint like soc at the begining to get some good features out of the data

based on these: Fewer gates. Under gate noise, every gate costs accuracy, and two-qubit gates cost

twice. Try fewer feature-map repetitions or ansatz layers, linear instead of full entanglement, merged single-qubit rotations, simpler encodings, fewer qubits, and removing the gates that cannot influence the readout (the notebook shows that 12 of the 32 parameters of the baseline never reach the measured qubit).

 Better readout and use of the data: other readout qubits or observables, class weights instead of undersampling, data re-uploading [19], and an initialisation that keeps gradients large [13, 14].

 Noise-aware training: train inside the noise model [16] and compare with training without noise and deploying under noise. Noise also flattens the gradients of deep circuits [15].

 Cheaper training: validate less often or on a smaller stratified subset, stop once a target validation AUPRC or loss is reached, use SPSA [11], shot-frugal or stochastic gradient estimates [8, 9, 10], or layerwise training [12]. Show how close the cheap model stays to the fully trained one, in AUPRC and in its ranking of the transactions.

 Error mitigation: zero-noise extrapolation [17], and what it costs in circuit executions.

 Further afield: classical surrogates of the trained circuit [20], or truncated simulation with Pauli propagation [21]

Where could we improve the vqc? What degrees of freedome could we influence for better performance.

Explain the vqc to me so I have a better grasp upon it

Can you make a run grid for the two models to train them exactly as the task 2 ask for from the task1 guided?

putem sa facem cumva sa antrenam encoderul ala previously cumva mai bine sa clasifice datele de frauda separat de cele de legite cumva si apoi astfel vqc ul v a fi mai usor pt el?

ok, si dac apastram asta si incercam o implementare de algoritmi genetici pe vqc, cum ar merge?